3.1 Spot the Flaw

WORKSHEET

The question

STEP I 1991, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q6

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself When you are asked to "criticise each step", what should you check at each step? Make a list before you start.

STAGE 1 Is the derivative right?

Ask yourself Differentiate g with the quotient rule. Where is $g'(x) = 0$? Check $x = 1$ and $x = -1$ directly.

STAGE 3 Does comparing values tell you which is a maximum?

Ask yourself Suppose a function has exactly two stationary points, with values $\frac{3}{4}$ and 1. Does it follow that the first is a minimum and the second a maximum?

STAGE 4 Do stationary values bound a function?

Pause and try Even with correct stationary points, why can't you conclude that every value of g lies between the smallest and largest stationary values? Find a value of x with $g(x) < \frac{3}{4}$.

STAGE 5 The integral

Ask yourself Differentiate $-3(1-x)^{-4}$. Do you get $(1-x)^{-3}$?

Ask yourself With the correct antiderivative, $[\frac{1}{2}(1-x)^{-2}]_{-1}^{3} = \frac{1}{2}\cdot\frac{1}{4} - \frac{1}{2}\cdot\frac{1}{4} = 0$. So the answer was right after all. Or was it?

Try this next

1 **A substitution that lies.** (From STEP I, 1987.) Explain why the substitution $x = 1/t$ does not show that

$$\int_{-1}^{1} \frac{\mathrm{d}x}{(1+x^2)^2} \quad \text{and} \quad \int_{-1}^{1} \frac{-t^2}{(1+t^2)^2}\,\mathrm{d}t$$

are equal. Evaluate both integrals correctly. (One is positive and one is negative, so

they certainly differ.)

2 **True or false?** (From STEP I, 1998.) Justify your answers.

(a) $a^{\ln b} = b^{\ln a}$ for all $a, b > 0$.

(b) $\cos(\sin\theta) = \sin(\cos\theta)$ for all real θ.

(c) There is a polynomial P such that $|\mathrm{P}(\theta) - \cos\theta| \leq 10^{-6}$ for all real θ.

(d) $x^4 + 3 + x^{-4} \geq 5$ for all $x > 0$.

3 **All horses are the same colour.** Find the flaw in this "proof" by induction: any one horse is the same colour as itself; given $n+1$ horses, the first n are the same colour by the inductive hypothesis, and so are the last n; the two groups overlap, so all $n+1$ are the same colour.

4 **An integral that does exist.** Show that $\int_{-1}^{1} |x|^{-1/2}\,\mathrm{d}x$ exists, although the integrand is undefined at 0, and find its value.

Further reading

- **E. A. Maxwell, *Fallacies in Mathematics*** (Cambridge University Press). A classic collection of plausible wrong arguments, organised by topic, written for sixth-formers.
- **Bernard Gelbaum & John Olmsted, *Counterexamples in Analysis*** (Dover). Hundreds of examples showing why the hypotheses of theorems matter. Dip in once you have started an analysis course.
- **Imre Lakatos, *Proofs and Refutations*** (Cambridge University Press). A brilliant dialogue about how mathematicians find, and fix, errors in proofs.

The full guided solution, and what the question leads to at university, is in Section 3.1:
https://beyondstep.keithmcnulty.org/ch03/spot-the-flaw.html